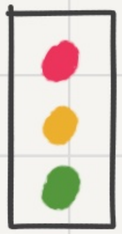
Probability

A random phenomenon is a situation in which we know the possible outcomes, but we don't know which outcome will occur.
We conduct trials (observations) to determine outcomes.

The sample space is the set of all outcomes.

Ex. •



$S = \{\text{red}, \text{yellow}, \text{green}\}$

We consider possible outcomes (or combinations of outcomes): events.

$E = \{\text{green}\}$

$F = \{\text{yellow}\} \cup \{\text{green}\}$

"union" [or]

• We flip a coin twice.

Let H denote heads; T denote tails.

$S = \{HH, HT, TH, TT\}$

Let's describe the events that:

E → we get at least one H;

F → we get at least one T.

$E = \{HH, HT, TH\}$

$F = \{HT, TH, TT\}$

We can consider the "intersection": [and]

$E \cap F = \{HT, TH\}$.

If two events have no intersection, that is $E \cap F = \emptyset$ [empty], then E and F are mutually exclusive.

How do we define probability?

Law of Large Numbers.

If we repeat a process* over and over, the proportion of times an event occurs approaches a fixed value: this is the probability of the event.

* assume independence!

Probability is defined in terms of a long-run relative frequency:

$$P(E) = \lim_{n\to\infty} \frac{n(E)}{n}.$$

← # of times E occurs

← # of experiments

Q. You flip a coin 10 times, and you see 10 heads in a row. Does the coin "owe" you a tails? NO!

Theoretical Probability

Cardano

Fermat

Pascal

For events with equally likely outcomes:

$$P(E) = \frac{\#\text{ of outcomes in } E}{\#\text{ of possible outcomes}}.$$

Ex. Find the probability of drawing a face (J, Q, K) from a standard deck.

$$P = \frac{\#\text{ face cards}}{\#\text{ cards}} = \frac{12}{52}$$

Axioms of Probability

Consider an experiment with sample space S.
Let $E \subseteq S$ "subset" be an event. We define a number $P(E)$ that satisfies the following:

1. $0 \leq P(E) \leq 1$
2. $P(S) = 1$
3. If $E_1, E_2, E_3 \ldots$ are mutually exclusive,

$$P(E_1 \cup E_2 \cup E_3 \cup \cdots) = P(E_1) + P(E_2) + P(E_3) + \cdots$$

Here, $P(E)$ is the probability of the event E.

Given any event E, we define the complement of E, denoted E^c — this is the event that E does NOT happen.

Q. What is $P(E^c)$?

② $P(S) = 1 \Rightarrow P(E \cup E^c) = 1$.

③ Since E, E^c are mutually exclusive

$$P(E \cup E^c) = P(E) + P(E^c).$$

Thus, $P(E) + P(E^c) = 1$

$$\Rightarrow P(E^c) = 1 - P(E).$$

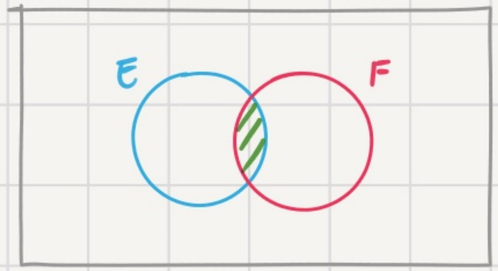
Some Rules of Probability

(Addition) If E, F are mutually exclusive,

$$P(E \cup F) = P(E) + P(F).$$

(General Addition) Given two events E, F:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$



* If E, F are mutually exclusive, $P(E \cap F) = 0$.

Ex. In a class of 30 students, 7 got an A in chemistry, and 11 got a B.

→ The events "getting an A" and "getting a B" are mutually exclusive.

e.g. $P(A) = 7/30$, $P(B) = 11/30$

$P(A \cap B) = 0$, $P(A \cup B) = \frac{7+11}{30}$.

Ex.

	Grades	Internships	CCA	
Boys	117	50	60	227
Girls	130	91	30	251
	247	141	90	478

- $P(\text{girl}) = \frac{247}{478}$
- $P(\text{girl and internship}) = \frac{91}{478}$
- $P(\text{CCA}) = \frac{90}{478}$
- $P(\text{CCA} \mid \text{girl}) = \frac{30}{251}$
- $P(\text{girl} \mid \text{CCA}) = \frac{30}{90}$

Let E, F be two events. We define the conditional probability of E given F denoted by $P(E \mid F)$, as the prob. that E happens taking into account the condition F.

$$\boxed{P(E \mid F) = \frac{P(E \cap F)}{P(F)}}$$

(General Multiplication) Given events E, F:

$$P(E \cap F) = P(E \mid F) \times P(F).$$

Two events E, F are independent if the occurence of one event does not affect the occurence of the other:

$$P(E \mid F) = P(E).$$

(Multiplication). If E, F are independent,

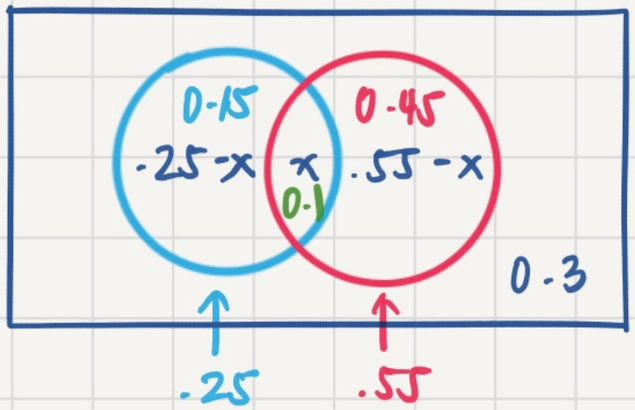
$$P(E \cap F) = P(E) \times P(F).$$

⚠ Mutually exclusive $\neq$ Independent !

Ex. In a certain city:

- 25% of residents own a cat;
- 55% own a dog;
- 30% do not own pets.

Find: $P(\text{cat} \mid \text{dog})$,
$P(\text{no dog} \mid \text{cat})$.



$$(.25 - x) + x + (.55 - x) = .7 \Rightarrow x = 0.1$$

$$P(C \mid D) = \frac{P(C \cap D)}{P(D)} = \frac{0.1}{0.55}$$

$$P(D^c \mid C) = \frac{P(D^c \cap C)}{P(C)} = \frac{0.15}{0.25}$$

Bayes' Theorem

By the definition of conditional prob.:

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}$$

$$P(F \mid E) = \frac{P(E \cap F)}{P(E)}$$

$$\boxed{P(E \mid F) = \frac{P(F \mid E)\, P(E)}{P(F)}}$$

Ex. Based on a survey:

- 71% of students use 𝕁,
- 18% of students use Instagram,
- 15% use both.

→ Are 𝕁 and I mutually exclusive?

→ Are they independent?

Mutually Exclusive: $P(E \cap F) = 0$

Independent: $P(E \mid F) = P(E)$

Use a table:

	𝕁	no 𝕁	Total
I	0.15	0.03	0.18
no I	0.56	0.26	0.82
Total	0.71	0.29	1.00

- $P(\text{𝕁} \cap I) = 0.15 \neq 0$

  $\Rightarrow$ not mutually exclusive!

- $P(I \mid \text{𝕁}) = \dfrac{0.15}{0.71} \approx 0.21 \leftarrow \dfrac{P(I \cap \text{𝕁})}{P(\text{𝕁})}$

  vs. $P(I) = 0.18 \neq P(I \mid \text{𝕁})$

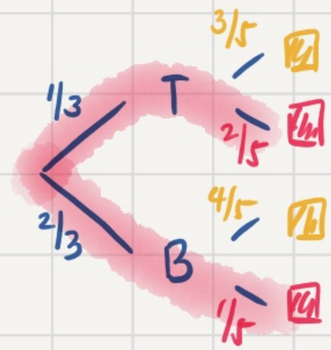
  $\Rightarrow$ not independent!

Ex.

Suppose that you are twice as likely to select a can from the lower shelf.

- Find P(tomato).
- You see that you got tomato soup. What is the probability that it came from the top shelf?

Construct a tree:



We have $P(B) = 2P(T)$.

Since $P(T) + P(B) = 1$

$P(T) + 2P(T) = 1$.

Thus, $P(T) = 1/3$.

- $P(\text{tomato}) = P(\text{tomato} \cap T) + P(\text{tomato} \cap B)$

  $= \frac{1}{3} \cdot \frac{2}{5} + \frac{2}{3} \cdot \frac{1}{5} = \frac{4}{15}$

- $P(T \mid \text{tomato}) = \dfrac{P(T \cap \text{tomato})}{P(\text{tomato})} = \dfrac{\frac{1}{3} \cdot \frac{2}{5}}{4/15} = \dfrac{1}{2}$